Paper III — Final Internal Audit Report

Linear-Error Clique Partitions of Split Graphs (Erdos #81)

OVERALL VERDICT: ALL AUDITED BLOCKS PASS

1. Scope

This report consolidates the independent internal audit of the finite and closed-form claims of Paper III. Every claim was re-derived from scratch, independently of the manuscript's own scripts and of the Lean 4 / Mathlib formalization. Numeric grid checks use exact rational arithmetic; algebraic identities are proved symbolically with SymPy; fractional optima use direct linear programming (HiGHS); integral triangle-packing numbers use exact 0/1 integer programming (CBC). The two external asymptotic inputs AX1 (Haxell-Rodl/Yuster) and AX2 (Dross + Barber-Kuhn-Lo-Osthus) are the paper's declared axioms and are outside the scope of computational audit.

2. Results by block

Block 01 — Algebraic identities

VERDICT: 12/12 identities PASS

Method: SymPy exact symbolic proof (simplify(LHS-RHS)=0 / exact rational / sum-of-squares) of the T(G) key identity (Thm 4.2), the (9.12) and (9.20) coefficients, the (9.19) completed square and its lower bound, delta>=7/8 for both parities (9.10), the corridor threshold p=2304, mu continuity at alpha=2/3, and the (4.5) closed forms.

results SHA-256: 7d5ea305ff7f44013184d2187d7e310efde926881cf24a4424e55667dbfe8519
zip SHA-256: eaf14d9dfbe6871a0dae73902fedf2aabacd2b31fb13df0b3e63b5c1c7c7cae7

Block 02 — Common-profile LP (Theorem 3.1 / E-3.1)

VERDICT: 351/351 instances PASS (max |LP-F| = 3.9e-14)

Method: Direct fractional triangle-packing LP (SciPy HiGHS) solved on the actual graph H(p,q,d) over the grid 3<=p<=8, 0<=q<=8, 0<=d<=p, and compared against the closed form F(p,q,d). Independent of the reduced 4-variable LP used in the proof.

results SHA-256: 7fabebbc8a96ff39e2468239a98f8cf3de2d5d0642ef6382391bf4628cc3b509
zip SHA-256: 6afed288c0d5beed04881546eac92fa8b47be3302f923a0450206f93067dcd39

Block 03 — Unified fractional margin (Theorem 4.2 / E-4.2)

VERDICT: 78,384/78,384 exact-rational checks PASS

Method: Exact-rational grid audit (fractions.Fraction, no floating point) of the completion-of-squares inequality (4.5) over 3<=p<=48, 1<=q<=2p, 0<=d<=p, with third-branch dominance bookkeeping.

results SHA-256: 577450739c3652ed9be447a3b426f23f8e856dd7eac4aa37002bcdb859b5239a
zip SHA-256: 46a438c43896dcf67e6a210d887a9d66963cb7c8f4903076be5d1acdc9999fab

Block 04 — Corridor integral packing (Lemma 5.1 & Cor 5.3)

VERDICT: E-5.1 180/180, Cor 5.3 180/180, 372 instances PASS

Method: Exact 0/1 ILP (PuLP + CBC) computation of nu3(G) on 372 systematically generated split graphs, verifying E-5.1 and Corollary 5.3 on the applicable instances (q >= r_p) plus the basic invariants 0<=Phi and 3*nu3<=|E|.

results SHA-256: eeb55cbb2b33abcbeae841b6ed2562b1a3ba039c3dfdbe5f0074d2561b9fa7ce
zip SHA-256: 38ee6759a4a17ff486c7b691c607d7b8b32540969bd4aa4f6688d17ae232bdf3

3. Methodology & tooling

Python 3.14; SymPy 1.14 (symbolic identities); SciPy 1.17 HiGHS (fractional LP); PuLP + CBC (exact ILP); fractions.Fraction (exact rationals). Each block's verify_*.py writes its full log to results/ and returns nonzero on any failure. The per-block PDF certificates and zips are produced by build_audit_artifacts.py; this master report is produced by make_master_certificate.py.

4. Relationship to the formalization

This computational audit is complementary to the machine-checked Lean 4 / Mathlib development. The formalization certifies the full logical chain (Theorem 1.1 and the elementary core) relative to AX1 and AX2; this audit independently certifies the finite numeric and closed-form facts those proofs rely on. Agreement between the two is corroborating, not circular: they share no code.

Generated: 2026-07-21 16:41 UTC
Auditor: internal automated audit harness (Claude Code)
This certificate attests to computational and symbolic verification of the stated finite/closed-form claims at the audited parameter ranges.